ain.py

```
def encode(d):
    h = [0] * 7
    h[2] = d[0]
    h[4] = d[1]
    h[5] = d[2]
    h[6] = d[3]
    h[0] = h[2] ^ h[4] ^ h[6]
    h[1] = h[2] ^ h[5] ^ h[6]
    h[3] = h[4] ^ h[5] ^ h[6]
    return h
def detect_and_correct(h):
    c1 = h[0] ^ h[2] ^ h[4] ^ h[6]
    c2 = h[1] ^ h[2] ^ h[5] ^ h[6]
    c4 = h[3] ^ h[4] ^ h[5] ^ h[6]
    error_pos = c1 + (c2 * 2) + (c4 * 4)
    if error_pos != 0:
        print(f"Error detected at bit position: {error_pos}")
        h[error_pos - 1] ^= 1
        print("Error corrected successfully.")
    else:
        print("No error detected.")
    return error_pos
def to_str(bits):
    return "".join(map(str, bits))
def main():
    print("=== HAMMING CODE: ERROR DETECTION AND CORRECTION ===")
    n = int(input("Enter number of 4-bit data blocks: "))
    for b in range(1, n + 1):
        print(f"\n--- Block {b} ---")
        d = list(map(int, input(
            "Enter 4 data bits (e.g. 1 0 1 1): "
        ).split()))
        if len(d) != 4 or any(bit not in (0, 1) for bit in d):
            print("Invalid input! Enter exactly four bits (0 or 1).")
            continue
        h = encode(d)
        print("Original Data Bits:", to_str(d))
        print("Encoded Hamming Code:", to_str(h))
        pos = int(input(
            "Enter error position (1-7, 0 for no error): "
        ))
        if 1 <= pos <= 7:
            h[pos - 1] ^= 1
            print("Error introduced at position:", pos)
        elif pos != 0:
            print("Invalid position! No error introduced.")
        print("Received Code:", to_str(h))
        print("\nChecking for errors...")
        detect_and_correct(h)
        print("Corrected Code:", to_str(h))
        recovered_data = [h[2], h[4], h[5], h[6]]
        print("Recovered Data Bits:", to_str(recovered_data))
if __name__ == "__main__":
    main()
```

Scanned with OKEN Scanner

Output

```
=== HAMMING CODE: ERROR DETECTION AND CORRECTION ===
Enter number of 4-bit data blocks: 2

--- Block 1 ---
Enter 4 data bits (e.g. 1 0 1 1): 1 0 1 1
Original Data Bits: 1011
Encoded Hamming Code: 0110011
Enter error position (1-7, 0 for no error): 3
Error introduced at position: 3
Received Code: 0100011

Checking for errors...
Error detected at bit position: 3
Error corrected successfully.
Corrected Code: 0110011
Recovered Data Bits: 1011

--- Block 2 ---
Enter 4 data bits (e.g. 1 0 1 1): 0 1 1 0
Original Data Bits: 0110
Encoded Hamming Code: 1100110
Enter error position (1-7, 0 for no error): 6
Error introduced at position: 6
Received Code: 1100100

Checking for errors...
Error detected at bit position: 6
Error corrected successfully.
Corrected Code: 1100110
Recovered Data Bits: 0110
```

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